Solve with us 6.2 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1Key Points: 1

- Let $T: V \rightarrow W$T : V → W be a linear transformation. We define kernel of TT (denoted by $ker(T)$ ker(T)) and image of TT (denoted by $Im(T)$Im(T)) as follows:

$$Ker(T) = \{v \in V \mid T(v) = 0\}\text{Ker(T)} = \{\text{v} \in \text{V} \mid \text{T(v)} = 0\}$$
$$Im(T) = \{w \in W \mid \text{ there exists } v \in V \text{ for which } T(v) = w\}\text{Im(T)} = \{\text{w} \in \text{W} \mid \text{ there exists v} \in \text{V for which T(v)} = \text{w}\}$$

- TT is injective (i.e., TT is a monomorphism) if and only if $Ker(T) = \{0\}$Ker(T) = {0}.
- A linear transformation $T: V \rightarrow W$T : V → W is surjective (i.e., TT is epimorphism) if and only if $Im(T) = W$Im(T) = W.
- $Nullity(T) = dim(Ker(T))$Nullity(T) = dim(Ker(T)) and $Rank(T) = dim(Im(T))$ Rank(T) = dim(Im(T)).
- Let $T: V \rightarrow W$T : V → W be a linear transformation. Then

$$Rank(T) + Nullity(T) = dim(V).\text{Rank(T)} + \text{Nullity(T)} = \text{dim(V)}.$$

This is known as the rank nullity theorem.

- Let $T: V \rightarrow W$T : V → W be a linear transformation. TT is injective if and only if $Nullity(T) = 0$ Nullity(T) = 0. TT is surjective if and only if $Rank(T) = dim(W)$Rank(T) = dim(W).

Consider the following linear transformation:

$T: R^3 \rightarrow R^2$ T : R3 → R2
$T(x, y, z) = (2x + 3z, 4y + z)$ T(x, y, z) = (2x + 3z, 4y + z)

Answer the following questions:
1 point
Which of the following Options is true?

[Hint: $T(x, y, z) = (2x + 3z, 4y + z) = (0, 0)$T(x, y, z) = (2x + 3z, 4y + z) = (0, 0), i.e., $2x + 3z = 0$ 2x + 3z = 0 and $4y + z = 0$4y + z = 0]

- ○ $ker(T) = \{(3y, y, -4y) \mid y \in R\}$ker(T) = {(3y, y, −4y) | y ∈ R}
- ○ $ker(T) = \{(6y, y, 4y) \mid y \in R\}$ker(T) = {(6y, y, 4y) | y ∈ R}
- ○ $ker(T) = \{(6y, y, -4y) \mid y \in R\}$ker(T) = {(6y, y, −4y) | y ∈ R}
- ○ $ker(T) = \{(3y, y, 4y) \mid y \in R\}$ker(T) = {(3y, y, 4y) | y ∈ R}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$ker(T) = \{(6y, y, -4y) \mid y \in R\}$ker(T) = {(6y, y, −4y) | y ∈ R}

1 point
Which of the following is a basis of $Ker(T)$Ker(T)?

- ○

$\{(3, 1, -4)\}\{(3, 1, -4)\}$

$\{(6, 0, -4), (0, 1, -4)\}\{(6, 0, -4), (0, 1, -4)\}$

$\{(6, 1, -4)\}\{(6, 1, -4)\}$

$\{(3, 0, -2), (0, 1, -4)\}\{(3, 0, -2), (0, 1, -4)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(6, 1, -4)\}\{(6, 1, -4)\}$

Find the nullity of TT.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1

1 point

Find the rank of TT.

[Hint: use rank nullity theorem.]

No, the answer is incorrect.
Score: 0
Feedback:
– Here dim(V) = 3 as V = R^3.
Accepted Answers:
(Type: Numeric) 2

1 point

1 point

Which of the following is true?

TT is both injective and surjective.

TT is injective but not surjective.

TT is surjective but not injective.

TT is neither injective nor surjective.

No, the answer is incorrect.
Score: 0
Accepted Answers:
TT is surjective but not injective.

1 point

Which option represents the kernel and image of the following linear transformation?

$T: R^2 \rightarrow R^2$ T : R2 → R2

$T(x, y) = (x, 0)$T(x, y) = (x, 0)

$ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(1, 0)\}$ker(T) = Span{(1, 0)}, Im(T) = Span{(1, 0)}.

○ $ker(T) = Span\{(1,0)\}, Im(T) = Span\{(0,1)\}$ker(T) = Span{(1, 0)}, Im(T) = Span{(0, 1)}.

○ $ker(T) = Span\{(0,1)\}, Im(T) = Span\{(1,0)\}$ker(T) = Span{(0, 1)}, Im(T) = Span{(1, 0)}.

○ $ker(T) = Span\{(0,1)\}, Im(T) = Span\{(0,1)\}$ker(T) = Span{(0, 1)}, Im(T) = Span{(0, 1)}.

No, the answer is incorrect.
Score: 0
Feedback:
• Ker(T) = {(0, y) | y ∈ R}.
Accepted Answers:
$ker(T) = Span\{(0,1)\}, Im(T) = Span\{(1,0)\}$ker(T) = Span{(0, 1)}, Im(T) = Span{(1, 0)}.

Key Points: 2Key Points: 2

- Let $T: V \rightarrow W$T : V → W be a linear. Follow the steps below to find bases for the null space and the range space of TT.

Step 1: Find the matrix AA corresponding to TT with respect to some standard ordered bases $\beta = \{v_1, v_2, ..., v_n\}$β = {v1, v2, ..., vn} and $\gamma = \{w_1, w_2, ..., w_m\}$γ = {w1, w2, ..., wm} for VV and WW respectively.
Step 2: Use row reduction on AA to obtain the matrix RR which is in reduced row echelon form.

Step 3: The basis of the solution space of $Rx = 0$Rx = 0 is the basis of null space of matrix AA and can be obtained by finding the pivot and non-pivot columns (dependent and independent variables) as seen earlier.

Step 4: The vectors $\begin{bmatrix} c_{11} \\ c_{12} \\ \vdots \\ c_{1n} \end{bmatrix}, \begin{bmatrix} c_{21} \\ c_{22} \\ \vdots \\ c_{2n} \end{bmatrix}, ..., \begin{bmatrix} c_{k1} \\ c_{k2} \\ \vdots \\ c_{kn} \end{bmatrix}$ $\begin{bmatrix} & & & \end{bmatrix}$c11c12...c1n , $\begin{bmatrix} & & & \end{bmatrix}$c21c22...c2n , ..., $\begin{bmatrix} & & & \end{bmatrix}$ck1ck2...ckn form a

basis of the null space of AA precisely when the vectors $v_1', v_2', ..., v_k' \in ker(T)$v1′, v2′, ..., vk′ ∈ ker(T) where $v_i' = \sum_{j=1}^{n} c_{ij} v_j$ vi′ = ∑j=1n cijvj, form a basis for ker(T)ker(T). Use the basis obtained in step 3 to thus get a basis for ker(T)ker(T).

Step 5: Recall that if $i_1, i_2, \ldots, i_r$ i1, i2, … , ir are the columns of RR containing the pivot elements, then the same columns of AA form a basis for the column space of AA.

Step 6: The vectors $\begin{bmatrix} d_{11} \\ d_{12} \\ \vdots \\ d_{1m} \end{bmatrix}, \begin{bmatrix} d_{21} \\ d_{22} \\ \vdots \\ d_{2m} \end{bmatrix}, ..., \begin{bmatrix} d_{r1} \\ d_{r2} \\ \vdots \\ d_{rm} \end{bmatrix}$ $\begin{bmatrix} & & & \end{bmatrix}$d11d12...d1m , $\begin{bmatrix} & & & \end{bmatrix}$d21d22...d2m , ..., $\begin{bmatrix} & & & \end{bmatrix}$dr1dr2...drm

form a basis of the column space of AA precisely when the vectors $w_1^{'}, w_2^{'}, ..., w_r^{'} \in im(T)$w1′, w2′, ..., wr′ ∈ im(T) where $w_i^{'} = \sum_{j=1}^{m} d_{ij} w_j$ wi′ = ∑j=1m dij wj, form a basis for $im(T)$im(T). Use the basis obtained in step 5 to thus get a basis for $im(T)$im(T).

Consider the following linear transformation:

$T: R^3 \rightarrow R^3$ T : R3 → R3

$T(x, y, z) = (x - z, 2x + 3y + z, 3y + 3z)$ T(x, y, z) = (x − z, 2x + 3y + z, 3y + 3z)

Answer the following questions 7-9 using the information given above.

1 point

Which of the matrices corresponds the given linear transformation TT with respect to the standard ordered basis of R^3 R3 for both the domain and the codomain?

○ $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 3 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$

○ $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 3 \\ 0 & 1 & 0 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$

1 point

What will be the kernel of TT?

○ $\{(x, 0, z), (0, y, z) \mid x, y, z \in R\}$ {(x, 0, z), (0, y, z) | x, y, z ∈ R}.

○ $\{(z, 0, z), (0, -z, z) \mid z \in R\}$ {(z, 0, z), (0, −z, z) | z ∈ R}.

○ $\{(z, -z, z) \mid z \in R\}$ {(z, −z, z) | z ∈ R}.

○ $\{(x, 0, z), (0, -x, z) \mid x, z \in R\}$ $\{(x, 0, z), (0, -x, z) \mid x, z \in R\}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(z, -z, z) \mid z \in R\}$ $\{(z, -z, z) \mid z \in R\}$.

1 point

What will be the basis of the image space of T T found by the algorithm mentioned in the Key Points ?

○ $\{(1, 0, -1), (2, 3, 1)\}$ $\{(1, 0, -1), (2, 3, 1)\}$

○ $\{(1, 0, 0), (0, 1, 0)\}$ $\{(1, 0, 0), (0, 1, 0)\}$

○ $\{(1, 2, 0), (0, 3, 3)\}$ $\{(1, 2, 0), (0, 3, 3)\}$

○ $\{(-1, 1, 3)\}$ $\{(-1, 1, 3)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(1, 2, 0), (0, 3, 3)\}$ $\{(1, 2, 0), (0, 3, 3)\}$

Check Answers

Your score is: 0/9